Why Domain Matters, part 1

Retail and grocery

The record says

`5 kg atta: sales −59% week on week.`

What actually happened

The shelf was empty for 39 open hours.

Monday 15 June, 9 a.m. Store 014’s weekly trading review.

5 kg atta sold 59% less than the week before. The AI assistant explained why.

> **AI assistant, given the warehouse schema**
>
> Sales fell 59%, from 217 bags to 88. The books showed 33 to 78 bags on hand all week, so stock was available and this is not a supply problem. 5 kg's share of atta fell from 40% to 25%, which suggests shoppers are moving to other pack sizes.

Every number in that answer is right. The conclusion is not.

Five readings of one week

Data product	What it reports	What it concludes
The pipeline	11 of 12 data tests pass; the failure is full-cream milk below zero	“Healthy.”
The dashboard	Sales −59%; on-shelf availability 97%, from book stock	“Demand fell, with stock on hand.”
The forecast	28-day mean: 24.3 → 20.2 bags a day, 13% above recorded sales	“Tune it down.”
The mix analysis	5 kg share of atta units: 40% → 25% in a week	“Give 1 kg more shelf.”
The AI assistant	Seven questions about the first half of June	One of seven right

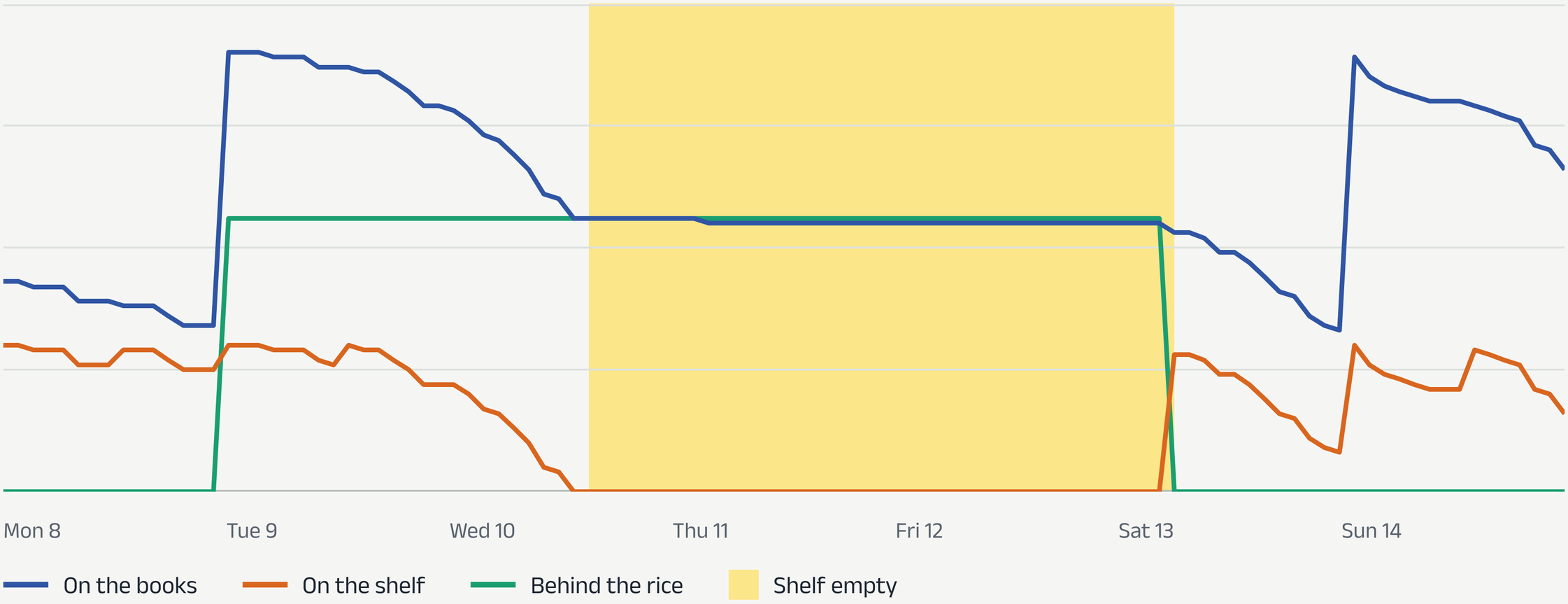
Each reads its inputs correctly. None of them can see the shelf.

What actually happened

Tue 9 Jun, 06:17	14 cases (56 bags) from the morning truck are put away behind the rice stack, in the wrong bay
Wed 10 Jun, 15:48	The shelf runs out. The books say 56 bags, above the reorder point of 40, so nothing is ordered
39 open hours	The shelf stands empty while the system shows stock
Sat 13 Jun, 10:00	Staff restacking the rice find the cases and fill the shelf
Shoppers	asked for 71 bags they could not find

Demand did ease after the month-start shop: shoppers came for 224 bags in the first week and 148 in the second. The rest of the drop was a shelf with nothing on it.

The books never fell to the reorder point



5 kg atta at Store 014, bags, every open hour from 8 to 14 June 2026. Grid lines every 25 bags. Synthetic world, seed 42.

Why the data could not say so

Sales are censored demand

A till records a sale, not a shopper. When the shelf is empty, sales measure the shelf, and every model trained on them inherits the gap: forecasts, elasticities, mix shares.

Book stock is a belief

It is computed: the last count plus receipts, minus sales and write-offs. It agreed with every transaction and was wrong about the shelf for three days. No data test looks at the shelf.

The holdout is censored too

Scored on recorded sales, the forecast looks 13% too high. Scored on the shoppers who came, it is 8% too low. The better a model fits the holdout, the worse it gets.

Conlon and Mortimer (2013) on demand estimation when availability changes; DeHoratius and Raman (2008) on inventory record inaccuracy.

None of this is rare

8.3%

of grocery items out of stock on average, in a worldwide study that traced most out-of-stocks to the store itself

65%

of nearly 370,000 inventory records wrong, across 37 stores of one retailer

5%

of open hours with an empty shelf across the world's seven warehouse items in June, the rate it was calibrated to

FMI, GMA and CIES (2002), worldwide out-of-stock study; DeHoratius and Raman (2008), Management Science 54(4); Store 014, June 2026.

Model how the data is observed

A gap detector learns each item’s hourly pattern from May’s till lines, adds up the sales it expects over every silence while the books show stock, and alerts once a silence that long is less likely than 1 in 1,000.

19:00

on Wednesday: the 5 kg gap flagged three open hours in

6 of 10

alerts in June at an empty shelf, across seven items; 6 of the 10 gaps found

June rerun, checking the shelf at every alert

Acting on	Gaps found	5 kg bags lost
Nothing, as today	–	157
Nightly rule, one day	2 of 10	90
Hourly, 1 in 100	7 of 10	33
Hourly, 1 in 1,000	6 of 10	35
Hourly, 1 in 10,000	3 of 10	46

The threshold is a business decision: a false alarm costs a walk to a full shelf; a missed gap costs bags.

Unconstrain demand before you train

20.2

bags a day: the store's 28-day forecast on 28 June, trained on recorded sales

25.6

bags a day, trained on unconstrained demand: the expected sales filled into every censored hour

24.9

bags a day that shoppers came for in June, from the answer key

Bias over June	Against recorded sales	Against shoppers
Store's forecast	+13%	−8%
On unconstrained demand	+22%	−1%

Scored on recorded sales, the better forecast looks worse. Order-up-to level at month end: 84 → 93 bags.

Give the assistant a domain layer

1 → 7 of 7

questions answered right, schema only → with the domain layer

- Definitions: “sales measure demand only while the item is on the shelf”; “book stock is a belief, not an observation”
- Three rules, among them: treat a silence the item’s usual rate cannot explain as censored
- The tables that observe the shelf: till lines, shelf audits, receipts, counts
- Four views: hourly sales, the hourly book, censored hours, a demand estimate

Why did 5 kg atta sales fall? With the domain layer:

From Wed 10 Jun, 15:53 to Sat 13 Jun, 10:43 the tills recorded one bag in 39 open hours, against 68 expected, while the books showed 55–56 bags. The shelf audit at Fri 12 Jun, 11:00 found the bay empty. No delivery arrived before sales resumed, so the bags were in the store but not on the shelf: misplaced or not refilled. With those hours filled in, the week comes to about 155 bags against 217: the empty shelf explains about half of the fall, and the rest is demand easing after a strong first week. Nothing says shoppers changed packs.

BIRD (Li and others, 2023): best model 40% execution accuracy, people 93%. Sequeda, Allemang and Jacob (2023): GPT-4 16% on the SQL schema, 54% on a knowledge graph.

One name, five numbers

97%

On the books

Open hours with stock on the books

75%

Shelf audits

8 audits, Tuesdays and Fridays at 11:00

79%

Books and tills

Stock on the books, outside the gaps the tills show

77%

On the shelf

Open hours with a bag on the shelf: answer key

78%

Demand served

Bags wanted that shoppers found: answer key

On-shelf availability of 5 kg atta in June. Whichever one your semantic layer encodes is the one the business will see.

Or change what gets recorded

Join the observation you already have

The 11:00 audit on Friday 12 June found the 5 kg bay empty while the books showed 55. The audit table was in the warehouse all along. Nothing joined it.

Capture the missing event

1 bag

lost in June instead of 157

Scan every case into a numbered backroom bin at put-away, and the record knows where the stock is. Sometimes the best model is a barcode scan.

Bags lost come from rerunning June in the world, on the same shoppers, with bin scans from 1 June.

Now let an agent act on it

A scripted replenishment agent reviews every warehouse item at 22:00. The same tools and the same rule: if three days' demand comes to less than half the forecast while the item was available, cut the forecast. One version reads the KPIs as reported. The other reads censored hours, and asks for a shelf check when a silence is too long.

No agent

157

bags of 5 kg atta lost this June; 221 on average over forty more

The store's ordering run alone

Reads the KPIs as reported

137

bags of 5 kg atta lost this June; 214 on average over forty more

Two forecast cuts, no shelf checks

Reads them with the domain rules

51

bags of 5 kg atta lost this June; 105 on average over forty more

Five shelf checks, two at an empty shelf

Scripted agents: rules standing in for a model's decisions. June rerun on the same shoppers; forty more Junes with different shoppers and the same failures.

Six questions to ask your team

1. Which of our tables are observations, and which are beliefs computed from other tables?
2. Where does our data stop recording what we are trying to measure?
3. Which independent observation do we have, and does anything join to it?
4. What do we score our models against, and does it share their blind spot?
5. What does our AI assistant know about how the business works, and how do we test that?
6. Before an agent acts on a KPI, what does it read, and what can it go and check?

For data and AI leaders. The technical companion answers each one for Store 014.

Sales are not demand. The book is not the shelf.

The tests in SQL, the models in Python, and a notebook on the records:

`kvinayakpai.github.io/domain-series/retail/technical`

> **Next: logistics**
>
> Delivered on time, in full. One pallet line held the wrong pack size.